Will the Next Word Switch Language? Anticipatory Code-Switch Prediction Across 15 Language Pairs

CS 6140 Machine Learning, Project Update 1 (Week 6, Fall 2026)

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Abstract

We predict code-switching before it happens: at every token of an utterance, using only the prefix, a model predicts whether the next token changes language and how long the new segment will last. This update covers data engineering, labeling, a leak-free streaming loader and the first baselines on all 15 language pairs of the SwitchLingua corpus (974,994 sentences). A logistic regression on prefix-only features reaches a macro switch F1 of 0.555 (PR-AUC 0.528) and a duration macro-F1 of 0.531, with large gaps between pairs. Code: https://github.com/a-ospan/streaming-code-switching.

1 Problem Description

Code-switching (CS), alternating between languages inside one utterance, is common among bilinguals, yet most NLP work identifies languages after the whole sentence is seen. Streaming speech recognition, predictive keyboards and dialogue agents instead need to *anticipate* a switch. We formulate anticipatory CS prediction: given only the prefix $w_1, \ldots, w_t$, a causal model predicts (i) $y_{sw} \in \{0, 1\}$, whether w_{t+1} is in a different language than w_t, and (ii) $y_{dur} \in \{1{-}2, 3{-}6, 7{+}\}$, the length in tokens of the new segment, defined only when $y_{sw} = 1$. We use the SwitchLingua text corpus [14] and evaluate the switch F1, the duration macro-F1, and *universality*: the standard deviation σ of switch F1 across language pairs, where lower means more consistent.

2 Background

Predicting switches. Solorio and Liu [9] were the first to learn where a speaker may switch, on a 40-minute Spanish–English conversation (about 8k words, 239 switches). They trained Naive Bayes and VFI classifiers on lexical, language and POS features of only the one or two preceding words, to suit real-time use. The best F1 was 0.28 with 96% negative instances, which shows how hard and imbalanced the task is. Chan et al. [3] ask whether word predictability drives switch points in 1,010 Cantonese–English switches from 41 bilinguals. They regressed switching on surprisal from autoregressive and masked language models. Less predictable words were switched more, consistently across bilingual groups. Xie et al. [13] predict upcoming switches token by token in Chinese–English dialogue (ASCEND). They compared windowed RNNs over BERT embeddings with fine-tuned mBERT and XLM-R. mBERT reached an AUC of 0.98, but on one language pair only.

Language identification in CS text. Burchell et al. [2] test whether scalable language identification can find CS text for corpus building. They recast it as sentence-level multi-label tagging and compared three systems on six CS test sets. Exact-match accuracy stayed below 0.16, so no current approach is adequate. Sterner [11] builds one word-level tagger that separates English from any other language. He merged five CS corpora and fine-tuned XLM-R with a named-entity head. It beat pair-specific systems by 2.3 to 4.6 points zero-shot on unseen pairs. Kargaran et al. [6] ask why sentence-level identifiers return only the dominant language of a CS sentence. Their training-free MaskLID masks words of the dominant language and re-runs the identifier. Detected CS rose up to 22-fold, yet one-word insertions were still mostly missed. Iakovenko and Hain [5] identify the matrix language that frames a CS utterance. They implemented Matrix Language Frame principles on text and trained audio classifiers on those labels for Mandarin–English and Spanish–English. Their best audio systems beat acoustic LID, but agreement between principles was sometimes low.

Benchmarks and models for CS. Aguilar et al. [1] built LinCE, a benchmark of 11 corpora over 4 language pairs and 4 tasks (LID, NER, POS, sentiment). They report LSTM, ELMo and mBERT baselines with a public leaderboard. It standardizes evaluation, but covers only four pairs. Khanuja et al. [7] built GLUECoS for Hindi–English and Spanish–English, with six tasks including QA and NLI. They compared cross-lingual embeddings with multilingual models, also fine-tuned on synthetic CS text. Multilingual models won, best when fine-tuned on CS data. Winata et al. [12] ask whether multilingual models handle CS. They compared them with bilingual embeddings and meta-embeddings on NER and POS for three pairs. Pretrained multilingual models did not guarantee good CS representations.

Multilingual backbones. Conneau et al. [4] train XLM-R, a masked language model, on more than two terabytes of text in 100 languages. It beat mBERT by 14.6 points on XNLI, most for low-resource languages, but it is bidirectional, so streaming use needs a causal mask. Xue et al. [15] pretrain mT5 on 101 languages and balance languages with exponent-smoothed sampling. It set the state of the art on many multilingual benchmarks; we reuse its sampling idea for unbalanced pairs. Lin et al. [8] train XGLM, left-to-right multilingual language models, and evaluate few-shot learning. Their largest model beat GPT-3 on multilingual reasoning and translation, making XGLM a natural causal backbone for us.

Our gap. Prior switch prediction covers one pair at a time, and CS benchmarks label switches after the fact. We predict switches and their duration from the prefix alone, on 15 pairs at once, and measure how consistent the predictor is across them.

3 Dataset

3.1 Corpus

SwitchLingua [14] contains 234,172 LLM-generated conversations with 19 columns (topic, speaker profile, CS type, quality scores). The

released text has **15 pairs**, more than the 12 languages in the paper: 7 non-Latin-script pairs with English (Arabic, Cantonese, Chinese, Hindi, Japanese, Korean, Russian), 7 Latin-script pairs with English (French, German, Italian, Malay, Filipino, Spanish, Vietnamese), and German–French. We pin version `6e048d27` of the corpus. Generation controls are balanced by design: 50% single-turn and 50% multi-turn conversations; three CS types (tag, inter-sentential, intra-sentential), each about a third; and six CS functions in equal shares. An LLM judge scored each row for fluency, naturalness, ratio and socio-cultural fit (mean 8.42 on a 10-point scale, range 5.0 to 9.75). A conversation holds 4.2 sentences on average, and a sentence 19.5 tokens (19.0M tokens in total). The pairs are very unbalanced: Cantonese–English is 64% of all sentences, while every other pair has 20k to 30k.

3.2 Pre-processing

Parsing. The text column is a stringified Python object: mostly lists of sentences, but also dictionaries (with topic, instances, speaker or message keys), nested lists and plain strings. Our parser handles all formats and keeps text fields only, giving 974,994 sentences; 5 empty or pasted-conversation sentences are removed and counted.

Tokenization. Words are split on whitespace and punctuation. Each Chinese character is one token, the mBERT convention. A whole run of characters as one token would leak the future: a run only ends when a non-Chinese character arrives, and 85% of the Cantonese runs that end glued to a letter end at a switch. Punctuation is not a token; it is kept as context for the preceding token.

Language ID resources. The corpus gives only each row's two languages, so we derive a language for every token (Section 3.3). For Latin-script pairs we use the `wordfreq` library [10], an offline snapshot (up to about 2021) of Wikipedia, subtitle, news, book, web and social-media text that gives each word a Zipf frequency per language. 56% of Hindi–English rows are romanized, so we built a romanized-Hindi lexicon (1,056 words) from the corpus itself, using the Latin words in Devanagari rows as the English reference. Hindi is therefore reported as two strata, giving 16 evaluation strata. As a sanity check, our per-row share of second-language tokens correlates with the ratio estimated by the corpus's own LLM judge (Pearson 0.47 to 0.60 for 13 pairs, 0.32 for Hindi and 0.29 for Filipino, on 300 rows per pair). Since that estimate is itself noisy, a hand-validation sample of 450 sentences (30 per pair) is prepared.

3.3 Labeling procedure

Every sentence goes through five steps; Table 1 shows the output of our code on two sentences.

(1) **Tokenize**: words, one token per Chinese character, one token per run of digits. Punctuation is not a token; it stays with the preceding token as context typed before the next one.
(2) **Script rule**: if the token's script belongs to only one of the row's two languages, it decides.
(3) **Frequency rule** (Latin-script pairs): the language whose `wordfreq` Zipf frequency is at least 0.3 higher wins; a word below 1.0 in both lists is unknown. Romanized Hindi rows use our lexicon.

Table 1: Labeling output of our code on two sentences: the rule that decided each token, its language and the targets. The comma after *2022* is part of the prefix at that token.

Token	Rule	Language	y_{sw}	y_{dur}
Spanish–English: "Vi a Messi en 2022, honestly the best game ever!"				
Vi	frequency	es	0	–
a	context (both lists)	es	0	–
Messi	context (unknown name)	es	0	–
en	frequency	es	0	–
2022,	context (number)	es	**1**	**3–6**
honestly	frequency	en	0	–
the	frequency	en	0	–
best	frequency	en	0	–
game	frequency	en	0	–
ever	frequency	en	–	–
Russian–English: "В прошлом году *Marvel* сняли фильм, *you know?*"				
В	script	ru	0	–
прошлом	script	ru	0	–
году	script	ru	**1**	**1–2**
Marvel	script (name)	en	**1**	**1–2**
сняли	script	ru	0	–
фильм,	script	ru	**1**	**1–2**
you	script	en	0	–
know	script	en	–	–

Table 2: Fixed splits, by conversation and stratified by stratum.

Split	Conversations	Sentences	Target positions
Train (80%)	187,329	778,867	14.4M
Validation (10%)	23,418	98,046	1.8M
Test (10%)	23,422	98,081	1.8M

(4) **Context rule**: tokens still undecided (numbers, unknown names, words frequent in both languages) take the language of the nearest decided token, preferring the left one.
(5) **Targets**: $y_{sw}(t) = 1$ if the language of token $t+1$ differs from token t; then $y_{dur}(t)$ is the bin (1–2, 3–6, 7+) of the run of same-language tokens starting at $t+1$. The last token and non-switch durations get no target (-1, ignored by the loss and the metrics).

Labels may use the whole sentence (the context rule can look right); baseline features and model inputs never do. Names get no special label: a name in the other script (*Marvel* in Russian text) or frequent in the other language's list counts as a switch, which the hand validation will quantify. Unit tests check every step on hand-worked sentences.

3.4 Splits

We split 80/10/10 *by conversation*, stratified by stratum with a fixed seed, and keep conversations sharing a sentence (5,586 duplicates) in one split (Table 2). Table 3 and Figure 1 summarize the labels. Switches are rare (5 to 13% of positions for most pairs), so we report F1 and PR-AUC rather than accuracy.

Table 3: Sentences and switch rate per stratum (all splits).

Stratum	Sentences	Switch rate
Chinese–English	29,561	5.1%
Japanese–English	27,893	6.3%
Cantonese–English	619,863	6.8%
Italian–English	24,165	8.4%
German–French	23,598	8.6%
Arabic–English	26,791	9.1%
Vietnamese–English	26,803	9.1%
Russian–English	24,625	9.3%
Spanish–English	24,517	9.6%
German–English	19,583	9.8%
French–English	23,563	10.3%
Korean–English	25,388	10.6%
Malay–English	26,127	13.3%
Hindi–English (Devanagari)	11,186	16.1%
Filipino–English	25,826	19.1%
Hindi–English (romanized)	15,505	31.4%

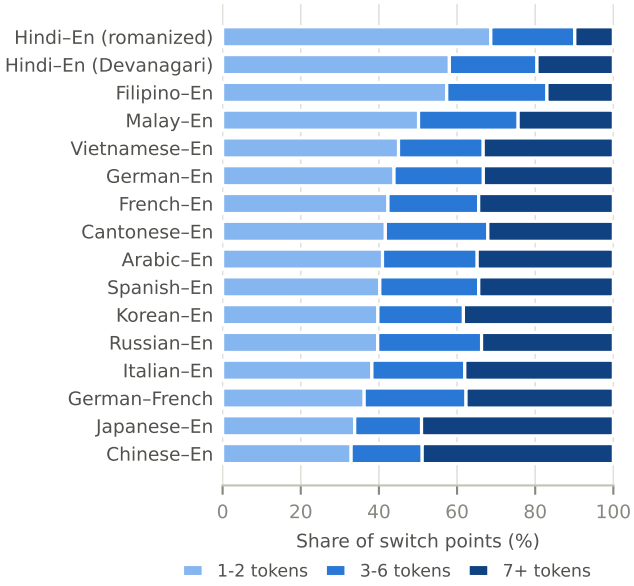


Figure 1: Duration class of the segment after a switch, per stratum. Hindi and Filipino are dominated by short insertions; Chinese and Japanese segments, counted in characters, are often 7+.

4 Experimentation and Results

4.1 Streaming setup

At token t the input is the prefix only. For transformers, each sentence is encoded once with the model's subword tokenizer and the prediction for word t is read at the last subword that ends before word $t+1$. A position is skipped if a subword spans the boundary or if the prefix tokenizes differently on its own, so no future character can leak; automated tests change every word after t and check that inputs at t do not change. To keep Cantonese from dominating training, pairs are drawn with probability proportional to $n_p^{0.3}$, as in mT5 [15], which lowers Cantonese from 64% to 16% of samples.

Table 4: Baselines on the validation split (macro average over 16 strata). σ: standard deviation of switch F1 across strata.

Model	F1	PR-AUC	P	R	σ	Dur. F1
Majority	0.000	0.114	0.000	0.000	n/a	0.202
Logistic reg.	**0.555**	**0.528**	0.550	0.571	0.076	**0.531**

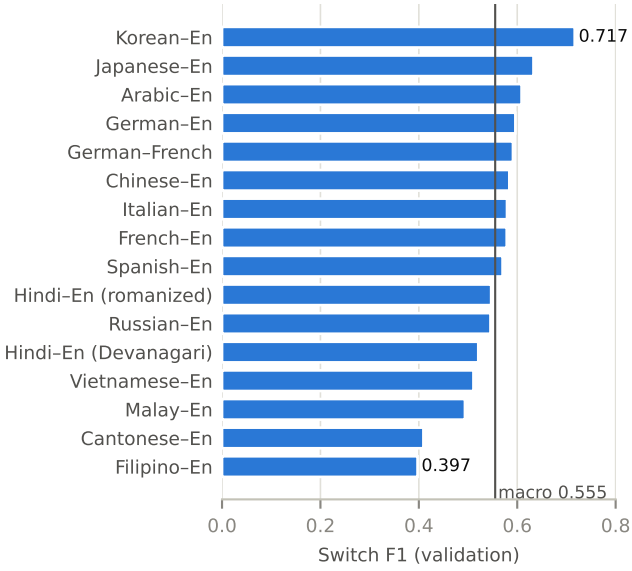


Figure 2: Logistic regression switch F1 per stratum (validation). The line is the macro average.

4.2 Baselines

Majority class always predicts no switch and the most frequent duration (1–2). **Logistic regression** uses prefix-only features: current language, length of the current run, switches so far, position, punctuation typed after the token, script, stratum, and stratum × current language. Tags are rebuilt from the prefix only, since the context rule may look right. A binary model predicts y_{sw} and a multinomial (softmax) model predicts y_{dur}, trained on true switch points. Both use 3,000 training sentences per stratum (838k positions); the switch threshold (0.225) is tuned on training data. All results are on the validation split (1.8M positions), macro-averaged over the 16 strata.

4.3 Results

Table 4 shows the logistic regression far above the majority class, with balanced precision and recall. Figure 2 shows large gaps between pairs: Korean (0.717) and Japanese (0.633) are easiest, Filipino (0.397) and Cantonese (0.409) hardest, giving $\sigma = 0.076$.

Why Korean is easy and Cantonese hard. The strongest feature is punctuation (Table 5). In Korean, Chinese and Japanese, 68 to 70% of switches follow a comma or sentence end, versus 2 to

Table 5: Share of positions right after punctuation (2,500 validation sentences per pair) and baseline switch F1.

Pair	Switches	Other positions	Switch F1
Korean–English	68.6%	2.4%	0.717
Japanese–English	69.9%	3.7%	0.633
Chinese–English	67.5%	3.4%	0.584
Cantonese–English	38.0%	5.5%	0.409

Table 6: Duration confusion matrix of the logistic regression at true switch points (validation, all strata).

True \ predicted	1–2	3–6	7+	Recall
1–2	**56,337**	3,011	5,777	86.5%
3–6	17,264	**4,737**	14,357	13.0%
7+	10,395	3,914	**31,844**	69.0%

Table 7: Usable prediction positions per tokenizer (lowest strata) and XGLM unknown-subword rate.

Stratum	XLM-R	XGLM	XGLM unknown
Chinese–English	68.9%	75.2%	0.6%
Japanese–English	78.4%	85.3%	0.2%
Cantonese–English	79.6%	85.3%	5.4%
Korean–English	99.9%	95.8%	0.6%

4% of other positions. In Cantonese only 38% do: English words are inserted mid-clause (呢個IG update好正), with no visible cue. The baseline thus learns clause-level switching but not insertions.

Duration. Short (1–2) and long (7+) segments are recognized well (87% and 69% recall), but only 13% of 3–6 segments are (Table 6), since prefix features cannot tell a phrase from a clause.

Tokenizer alignment. We checked how the streaming loader fits each backbone's tokenizer on 300 training sentences per stratum (Table 7). In Chinese-character text, 20 to 30% of character positions fall inside one XLM-R subword and are skipped, but no true switch point is lost. XGLM lacks common Cantonese characters (5.4% unknown subwords) and loses 4 to 5% of switch points in Chinese and Cantonese. All other strata keep at least 96% of positions with both tokenizers. Later model comparisons will therefore use positions valid for both models.

5 Plan for Next

Week 7: causal masking and the two prediction heads, already implemented with a test showing that later tokens cannot change earlier outputs. Weeks 8 to 10: multitask training of XLM-R with a causal mask [4] and the natively causal XGLM [8], tuning the loss weights, comparing sampling exponents 0.3 and 0, and comparing backbones against these baselines. Weeks 11 and 12: per-pair evaluation, universality σ and bootstrap confidence intervals, and hand validation of the language tags. Weeks 13 and 14: error analysis by script and switch type, the streaming demo and the final paper.

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